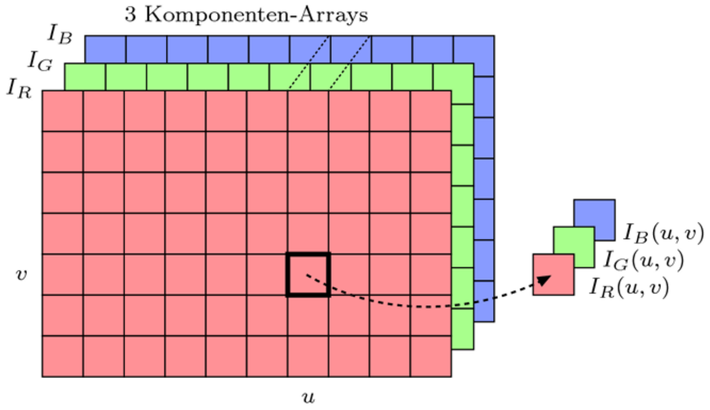
Image Representation in Memory/Files
Ordered by Components (RGB)

- „Planes“: Three compononets are separeted in 3 arrays of same size
- In memory or in files, these arrays are linear, of course



YUV (Component) Color Space

- Color space conversion: YUV -> RGB

$$Y = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B$$
$$U = 0.492 \cdot (B - Y)$$
$$V = 0.877 \cdot (R - Y)$$

- In matrix form (linear mapping)

  RGB -> YUV

$$\begin{pmatrix} Y \\ U \\ V \end{pmatrix} = \begin{pmatrix} 0.299 & 0.587 & 0.114 \\ -0.147 & -0.289 & 0.436 \\ 0.615 & -0.515 & -0.100 \end{pmatrix} \cdot \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

  YUV -> RGB

$$\begin{pmatrix} R \\ G \\ B \end{pmatrix} = \begin{pmatrix} 1.000 & 0.000 & 1.140 \\ 1.000 & -0.395 & -0.581 \\ 1.000 & 2.032 & 0.000 \end{pmatrix} \cdot \begin{pmatrix} Y \\ U \\ V \end{pmatrix}$$